Linux File System Explorer

Day 1 Investigation Worksheet — detailed responses

/etc

What did I ask?	What is the significance of the /etc directory in the Linux filesystem?
What did AI tell me?	The /etc directory stores system-wide configuration files used by the operating system and applications. These files define settings for services, networking, and other system components.
What did I expect?	A directory containing configuration files required to control system settings and application behavior.
What command/action did I perform?	ls -l /etc; ls -la /etc
What happened?	Listed the contents of /etc and displayed details such as permissions, ownership, file sizes, and modification times. Configuration files and directories related to system settings were visible.
Yes / No / Partially (AI correctness)	Yes
What did I learn?	/etc contains important system-wide configuration files. Modifying them can affect the operating system and its services, so changes should be made carefully.

/var

What did I ask?	What is the significance of the /var directory in the Linux filesystem?
What did AI tell me?	The /var directory stores variable data that changes frequently while the system is running, including logs, caches, application data, and other service-generated information.
What did I expect?	A directory used to store changing system data, particularly logs and application-generated files.
What command/action did I perform?	ls -l /var; ls -l /var/log
What happened?	Listed the contents of /var and /var/log. These locations can contain system data and log files useful for investigating errors and troubleshooting issues.
Yes / No / Partially (AI correctness)	Yes
What did I learn?	/var stores data that changes during system operation. The /var/log subdirectory is particularly useful for troubleshooting system events, application errors, and service failures.

/home

What did I ask?	What is the significance of the /home directory in the Linux filesystem?
What did AI tell me?	The /home directory is used to store users' personal files, documents, downloads, and user-specific settings. Each regular user typically has a separate home directory, such as /home/username.
What did I expect?	A directory used to store user-specific items, including files, directories, downloads, and personal configuration settings.

What command/action did I perform?	ls -l /home; echo $HOME; ls -l ~
What happened?	Listed the contents of /home, printed the current user's home-directory path with echo $HOME, and listed the contents of the current user's home directory using ls -l ~.
Yes / No / Partially (AI correctness)	Yes
What did I learn?	/home contains user-specific files and helps separate users' personal data and settings. The root user's home directory is generally /root rather than /home/root.

/tmp

What did I ask?	What is the significance of the /tmp directory in the Linux filesystem?
What did AI tell me?	The /tmp directory is used by users and applications to store temporary files needed during operations. Files may be removed during a reboot or through cleanup policies, depending on system configuration.
What did I expect?	A temporary storage location for users and applications to store files required for short-term operations.
What command/action did I perform?	ls -l /tmp
What happened?	Listed the contents of /tmp and observed the files and directories currently present in the temporary storage location.
Yes / No / Partially (AI correctness)	Yes
What did I learn?	/tmp stores temporary files needed for short-term operations. Files here should not be treated as permanent storage because cleanup policies may remove them automatically.

/usr

What did I ask?	What is the significance of the /usr directory in the Linux filesystem?
What did AI tell me?	The /usr directory stores many user-space applications, executable programs, libraries, documentation, and shared resources. Common subdirectories include /usr/bin, /usr/lib, /usr/share, and /usr/local.
What did I expect?	A directory containing installed programs, executable files, libraries, and supporting resources required by applications and system utilities.
What command/action did I perform?	ls -l /usr; ls -l /usr/bin; ls -l /usr/share
What happened?	Listed /usr and inspected selected subdirectories containing executable programs and shared resources used by applications and system utilities.
Yes / No / Partially (AI correctness)	Yes
What did I learn?	/usr contains much of the system's user-space software and supporting resources. Its subdirectories organize executables, libraries, documentation, and other shared data.

Final Learning

The investigation helped me understand the purpose of key Linux directories. Configuration files are stored in /etc, changing data and logs in /var, personal files in /home, temporary files in /tmp, and many user-space programs and shared resources

in /usr.